HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2024 Final Examination

Date: 20 December 2024 Time: 08:30–11:30

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **11** questions. You are allowed to bring an A4 double-sided, handwritten cheat sheet.
- Please write your name, student ID, email, lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all the questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$. Calculators are not allowed for the exam.

Problems	1	2	3	4	5	6	7	8	9	10	11	Total
Points	7	4	16	9	9	12	10	10	10	6	7	100
Score												

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Logic [7 points]

(a) **[3 points]** Let p, q, r denote primitive statements. Use the laws of logic to show that $p \to (q \vee r)$ is equivalent with $(p \wedge \neg q) \to r$.

(b) **[2 points]** Assume the domain to be all integers, $Prime(x)$ is a predicates that means x is prime. Express the following statement in terms of quantifiers, variable(s), predicates, and the inequality symbols $<$ or $>$.

For every integer n bigger than 1, there is a prime strictly between n and $2n$.

(Strictly between n and $2n$ means it can not be equal to n an it can not be equal to $2n$)

(c) **[2 points]** Negate the statement in (b) so that all negation symbols immediately precede predicates.

Answer: (a) Let p, q, r denote primitive statements. We aim to show that:

$$p \to (q \vee r) \equiv (p \wedge \neg q) \to r$$

1. $p \to (q \vee r)$
   $\equiv \neg p \vee (q \vee r)$ (Definition of implication)
2. $\equiv (\neg p \vee q) \vee (\neg p \vee r)$ (Distributive law)
3. Now consider the other side:
   $(p \wedge \neg q) \to r$
   $\equiv \neg(p \wedge \neg q) \vee r$ (Definition of implication)
4. $\equiv (\neg p \vee q) \vee r$ (De Morgan's Law)
5. $\equiv (\neg p \vee q) \vee (\neg p \vee r)$ (Since $(\neg p \vee q) \vee r$ is equivalent to $(\neg p \vee q) \vee (\neg p \vee r)$)

Thus, $p \to (q \vee r)$ is logically equivalent to $(p \wedge \neg q) \to r$.

(b)
$$\forall n \left((n > 1) \to \exists p \left(Prime(p) \wedge (n < p < 2n)\right)\right)$$

(c) $\exists n \left((n > 1) \wedge \forall p \left(\neg Prime(p) \vee (p \leq n \vee p \geq 2n)\right)\right)$

Problem 2: Sets and Functions [4 points]

(a) **[2 points]** Prove or disprove that if $(A \cup B) \subseteq (A \cap B)$, then $A = B$. Note that one approach can be showing that $A \subseteq B$ and $B \subseteq A$, which means that they have to be equal.

(b) **[2 points]** We know that any number of form $\sqrt[n]{p}$ is irrational when p is prime and n is an integer $n \geq 2$. How many irrational numbers of this form exist? Is it *finite*, *countable infinite* or *uncountable*? Briefly argue why.

(Note that $y = \sqrt[n]{p}$ means $y^n = p$)

Answer: (a) To prove $A \subseteq B$, assume $x \in A$. $x \in A \implies x \in A \cup B$ by the definition of union. Then This implies that $x \in A \cap B$ Because $(A \cup B) \subseteq (A \cap B)$. This means $x \in B$ according to the definition of intersection. Therefore, $A \subseteq B$.

The same argument can be applied for the other way. So we can prove that $A = B$

(b) Note that there are countably infinite prime numbers and countably infinite values for n. This implies that there are $|\mathbb{N}|^2$ many irrational numbers of this form, which is countable infinite.

Problem 3: Combinatorics and Probability [16 points]

(a) **[6 points]** Assume we pick four *different* numbers from $\{1, 2, \cdots, 100\}$. What is the probability that the difference between the largest and the smallest number is k? Note that $2 < k < 100$ and it is integer. Explain your solution.

(b) [**4 points**] There are three cards (A, B, C) in a box:

Card A: Both sides are red.

Card B: Both sides are black.

Card C: One side is red and one side is black.

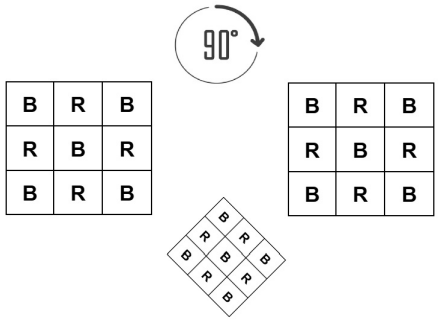
If we randomly select one card from the box and observe that one side is red, what is the probability that the other side is also red? Explain your solution.

(c) **[6 points]** Assume having a 3×3 grid carpet. We color each 1×1 tile with either Blue or Red randomly.

(i) **[3 points]** What is the probability that we get the same coloring if we rotate the carpet by 90° clockwise? Briefly explain your solution.
Hint: Think about which tiles need to have the same colors.

(ii) **[3 points]** What is the probability that we get the same coloring if we rotate the carpet by 180° clockwise? Briefly explain your solution.

As an example, the following picture shows a coloring that remains the same after a rotation of 90°.



Answer: (a) Without loss of generality, we can assume four numbers are $a < b < c < d$. We want to make sure that $d = a + k$. In order for this to happen, a has to be one of the values from 1 to $100 - k$. So there are $(100 - k)$ ways to pick a. by choosing a, the value for d will be fixed. So we need to pick two values b, c among $k - 2$ possible options that can be done in $\binom{k-1}{2}$ different ways.

The total possible ways to choose 4 values among 100 values is $\binom{100}{4}$. So the probability that the smallest and largest number differ by k can be calculated as:

$$p = \frac{(100 - k)\binom{k-1}{2}}{\binom{100}{4}}$$

(b) The probability that the other side is also red is the same as the probability that choosing card A. There are two possible "valid" options. And that is seeing either side of card A. And the total options are 3 cases. Either side of card A and the red side of card C. So the probability is $\frac{2}{3}$

(c) **For part i:** Note that if we want to have the same coloring after rotating the carpet by 90°, The corners must have similar colors which can have 2 cases.

The center can accept any color which has 2 options.

The middle tiles of the outer layer also must have the same color which can also accept 2 colors. So the total number of cases that the carpet have the same coloring after rotation is 2^3. The probability can be calculated as:

$$\frac{2^3}{2^9}$$

For part ii: Note that if we want to have the same coloring after rotating the carpet by 180°:

The top left and bottom right must have the same color which is 2 options.

The top right and bottom left must have the same color which is 2 options.

The center can accept any color which has 2 options.

The left and right tile of the center must have the same color which is 2 options.

The top and bottom of the center must have the same color which is 2 option.

So in 2^5 cases, the carpet remains the same after rotating 180°. So the probability is

$$\frac{2^5}{2^9}$$

Problem 4: Random Variable [9 points]

(a) **[3 points]** Assume X to be a random variable. Prove why $Var(aX) = a^2 Var(X)$.

Hint: You can try to use the following relation:

$$Var(X) = E(X^2) - E(X)^2$$

(b) **[6 points]** A Hash table is being used to store the phone numbers of k people, storing each person's phone number in a uniformly random location, represented by an integer between 1 and n.

(i) **[2 points]** Find the expected number of locations with no phone numbers stored.

(ii) **[2 points]** Find the expected number of locations with exactly one phone number.

(iii) **[2 points]** Find the expected number of locations with more than one phone number.
Hint: Should these quantities add up to n?

Answer: (a)

$$\begin{aligned} Var(X) &= E[(X - E[X])^2] \\ Var(aX) &= E[(aX - E[aX])^2] \\ &= E[(aX - aE[X])^2] \quad \text{(since } E[aX] = aE[X]\text{)} \\ &= E[a^2(X - E[X])^2] \\ &= a^2 E[(X - E[X])^2] \quad \text{(expectation is linear for constants)} \\ &= a^2 Var(X) \end{aligned}$$

Thus,

$$Var(aX) = a^2 Var(X)$$

(b) Let I_j be an indicator random variable equal to 1 if the jth location is empty, and 0 otherwise, for $1 \leq j \leq n$. Then $P(I_j = 1) = (1 - 1/n)^k$, since the phone numbers are stored in independent random locations. Then $I_1 + \cdots + I_n$ is the number of empty locations. By linearity of expectation, we have

(i)

$$E\left(\sum_{j=1}^{n} I_j\right) = \sum_{j=1}^{n} E(I_j) = n(1 - 1/n)^k.$$

(ii) Similarly, the probability of a specific location having exactly 1 phone number stored is $\frac{k}{n}(1 - \frac{1}{n})^{k-1}$, so the expected number of such locations is $k(1 - 1/n)^{k-1}$.

(iii) By linearity, the sum of the three expected values is n, so the expected number of locations with more than one phone number is $n - k(1 - 1/n)^{k-1} - n(1 - 1/n)^k$

Problem 5: Number Theory I [9 points]

(a) **[3 points]** Assume a to be an even integer. What are the possible values for $a^2 \pmod 8$?

Hint: Assume $a = 4k$ or $a = 4k + 2$.

(b) **[3 points]** Assume a to be an odd integer. What are the possible values for $a^2 \pmod 8$?

(c) **[3 points]** Assume a, b and c to be three integer numbers. Prove that the following equation has no integer solution.

$$a^2 + b^2 - 8c = 6$$

Hint: Try to use (a) and (b).

Answer: (a) If $a = 4k$ then $a^2 = 16k^2 \implies a^2 \equiv 0 \pmod 8$ If $a = 4k + 2$ then $a^2 = 16k^2 + 16k + 4 \implies a^2 \equiv 4 \pmod 8$. So if a is even then the possible remainder of $a^2 \pmod 8$ is either $\{0, 4\}$.

(b) $a = 2k+1 \implies a^2 = 4k^2+4k+1 = 4k(k+1)+1$. Note that $k(k+1)$ is always even. So $4k(k+1) \equiv 0 \pmod 8$. We can conclude that $a^2 \equiv 1 \pmod 8$.

(c) $a^2 + b^2 = 8c + 6$ Which means $a^2 + b^2 \equiv 6 \pmod 8$. From (a) and (b) we can see that the possible remainder of a square number mod 8 is $\{0, 1, 4\}$. So it is not possible to pick two values among this numbers such that the remainder mod 8 is equal to 6.

Problem 6: Number Theory II [12 points] Assume that a, b are integers and p is a prime number of form $4k + 3$ (with an integer k). As an example, 19 is a prime of this form while 17 is not of this form.

Assume that $p \mid a^2 + b^2$

(a) **[4 points]** Prove that if $p \mid a$ then $p \mid b$.

(b) **[8 points]** Prove that if $p \nmid a$ then there will be contradictions. (In other words, if $p \mid a^2 + b^2$ we can conclude that $p \mid a$ and $p \mid b$).

Hint: You might find little Fermat theorem useful here. You can also use part (a) even if you could not prove it.

Answer: (a) Note that if $p \mid a$ then $p \mid a^2$ which means $p \mid (a^2 + b^2) - a^2 \implies p \mid b^2 \implies p \mid b$.

(b) Assume that $p \nmid a$. This also implies that $p \nmid b$ (think why?). This means $gcd(p, a) = gcd(p, b) = 1$.

$p \mid a^2 + b^2 \implies a^2 \equiv -b^2 \pmod{p}$. Note that $p - 1$ is an even number and $\frac{p-1}{2}$ is odd.

$a^2 \equiv -b^2 \pmod{p} \implies (a^2)^{\frac{p-1}{2}} \equiv (-b^2)^{\frac{p-1}{2}} \implies a^{p-1} \equiv -b^{p-1}$.

By little Fermat theorem we know that $a^{p-1} \equiv b^{p-1} \equiv 1 \pmod{p}$. So this means $1 \equiv -1 \pmod{p} \implies 2 \equiv 0 \pmod{p}$. This is impossible as p is odd and this is the contradiction.

Problem 7 Chinese Remainder Theorem [10 points]

Solve the following system of congruences and show both steps:

1. Remove the coefficients of x.
2. Use Chinese Remainder Theorem to find the solution.

$$\begin{aligned} 3x &\equiv 4 \pmod 5 \\ x &\equiv 2 \pmod 4 \\ 3x &\equiv 6 \pmod 7 \end{aligned}$$

Answer: Note that the first step is to remove the coefficients.

Note that the inverse of 3 (mod 5) is 2. So if we multiply 2 by both sides of the first equation we have: $2 \cdot 3x \equiv 2 \cdot 4 \implies x \equiv 3 \pmod 5$

Note that as $gcd(3, 7) = 1$ we can simply divide both sides by 3 which gives us $x \equiv 2 \pmod 7$. So we need to solve the following system of congruences:

$$\begin{aligned} x &\equiv 3 \pmod 5 \\ x &\equiv 2 \pmod 4 \\ x &\equiv 2 \pmod 7 \end{aligned}$$

As all the modulos are coprime with each other. We can simply use chinese remainder theorem to find the solutions x.

Problem 8: RSA [10 points]

Consider an RSA system in which $p = 5$ and $q = 19$. Answer the following questions:

(a) **[2 points]** Can we choose $e = 9$? Why?

(b) **[4 points]** Use Extended Euclidean algorithm and calculate d when $e = 23$. Show all the steps.

(c) **[4 points]** Use repeated square method to calculate the encryption of $M = 3$. Show all the steps.

Answer: (a) Note that $n = 95$ $T = 4 \cdot 18 = 72$. e has to be co-prime with T but it is not. So it cannot be chosen.

(b)

$$\begin{aligned} 72 &= 3 \cdot 23 + 3 \\ 23 &= 7 \cdot 3 + 2 \\ 3 &= 1 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned}$$

$$\begin{aligned} 1 &= 3 - 1 \cdot 2 \\ &= 3 - 1 \cdot (23 - 7 \cdot 3) \\ &= 8 \cdot 3 - 1 \cdot 23 \\ &= 8 \cdot (72 - 3 \cdot 23) - 1 \cdot 23 \\ &= 8 \cdot 72 - 25 \cdot 23 \end{aligned}$$

Thus,

$$d \equiv -25 \pmod{72}$$

Since d must be positive, we add 72 to -25:

$$d = 72 - 25 = 47$$

(c)

$$\begin{aligned} 3^2 &\equiv 9 \pmod{95} \equiv 9 \\ 3^4 &\equiv 9^2 \pmod{95} \equiv 81 \pmod{95} \equiv 81 \equiv -14 \pmod{95} \\ 3^8 &\equiv 81^2 \pmod{95} \equiv 14^2 \pmod{95} \equiv 196 \pmod{95} \equiv 6 \pmod{95} \\ 3^{16} &\equiv 6^2 \pmod{95} \equiv 36 \pmod{95} \equiv 36 \end{aligned}$$

Since $23 = 2^4 + 2^2 + 2^1 + 2^0$:
We need to calcualte $3^{2^4} \cdot 3^{2^2} \cdot 3^2 \cdot 3 \equiv 72$

Problem 9: Algorithm and Time Complexity [10 points]

(a) **[5 points, 1 each]** For each pair of expressions (A, B), indicate whether A is O, Ω, Θ, of B. Note that zero, one, or more of these relations may hold for a given pair. **List all applicable relations. No explanation is needed.**

As an example for $A = n, B = 2n$, all O, Ω and Θ are correct.

(i) $A = \log n$ $B = 2n$

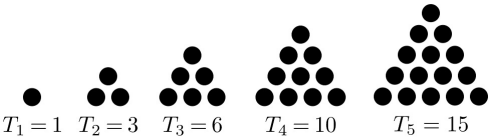
(ii) $A = 10n^2$ $B = 2n^2 + 10000n + 5n \log n$

(iii) $A = n^2$ $B = 5^{\log_2 n}$

(iv) $A = n^2 \log n$ $B = n \log^6 n$

(v) $A = n^3 (\log n)^5$ $B = n^{3.1}$

(b) Consider the following picture:



Consider the sequence of triangles formed by circles. Let $T(n)$ be the number of circles needed to form a triangle with n circles on each side.

(i) **[1 points]** Provide a recurrence relation for $T(n)$.

(ii) **[2 points]** Calculate the closed form formula for $T(n)$

(iii) **[2 points]** Prove the closed formula for $T(n)$ using induction.

Answer: (a) (i) O, Ω and Θ , (ii) $A = O(B)$, (iii) $A = \Omega(B)$, (iv) $A = O(B)$

(b) (i) $T(1) = 1, \quad T(n) = T(n-1) + n$

(ii) $T(n) = n + T(n-1) = n + (n-1) + T(n-2) = \cdots = n + (n-1) + (n-2) + \cdots + 1 = \frac{n \cdot (n+1)}{2}$

(iii) Base Case: $n = 1 : 1 = \frac{1 \cdot 2}{2}$.

Induction hypothesis: $n = k$ then $T_k = \frac{k \cdot (k+1)}{2}$

For $n = k + 1$ we need to prove $T_{k+1} = \frac{(k+1) \cdot (k+2)}{2}$.

$T_{k+1} = T(k) + (k+1) = \frac{k \cdot (k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$

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Problem 10: Recurrence [6 points]

Assume n is of form $n = 2^k$ for an integer k. Consider the following recurrence relation:

$$T(n) = 3T(\frac{n}{2}) + n \quad n > 1$$
$$T(1) = 1$$

Iterate the recurrence to get the closed form solution for $T(n)$.

Answer: Given:

$$T(1) = 1, \quad T(n) = 3T(n/2) + n \text{ for } n > 1$$

Using the repeated substitution method:

$$\begin{aligned} T(n) &= 3T(n/2) + n \\ &= 3(3T(n/4) + n/2) + n \\ &= 3^2T(n/4) + 3 \cdot \frac{n}{2} + n \\ &= 3^2T(n/4) + \frac{3n}{2} + n \\ &= 3^3T(n/8) + 3^2 \cdot \frac{n}{4} + 3 \cdot \frac{n}{2} + n \\ &= \cdots \\ &= 3^kT(n/2^k) + \sum_{i=0}^{k-1} 3^i \cdot \frac{n}{2^{k-i}} \end{aligned}$$

For $k = \log_2 n$:

$$T(n) = 3^{\log_2 n}T(1) + n \sum_{i=0}^{\log_2 n - 1} \left(\frac{3}{2}\right)^i = n^{\log_2 3} + n \sum_{i=0}^{\log_2 n - 1} \left(\frac{3}{2}\right)^i$$

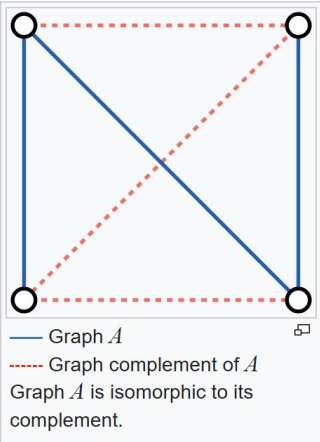
Since $\sum_{i=0}^{k-1} \left(\frac{3}{2}\right)^i$ is a geometric series:

$$\sum_{i=0}^{\log_2 n - 1} \left(\frac{3}{2}\right)^i = \frac{\left(\frac{3}{2}\right)^{\log_2 n} - 1}{\frac{3}{2} - 1} = \frac{n^{\log_2 \frac{3}{2}} - 1}{\frac{1}{2}} = 2(n^{\log_2 \frac{3}{2}} - 1)$$

Problem 11: Graph [7 points]

The complement or inverse of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G.

A self-complementary graph is a graph which is isomorphic to its complement. As an example, the following graph is self-complementary.



Assume a graph G is a self-complementary graph with n nodes. Consider H as its complement. Assume S to be the sum of all the nodes degrees in G.

(a) **[1 points]** Is S even or odd? No argument is needed.

(b) **[3 points]** Argue why $2S = n(n-1)$?
Hint: Two isomorphic graphs have same number of edges.

(c) **[3 points]** Prove that $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$.
You can use part (b) as a fact even if you could not solve it.

Answer: (a) Even

(b) Note that G and H are isomorphism, the sum of the degree of the nodes in these two graphs added is equal to $2S$. At the same time, note that for every node v, $d_v^G + d_v^H = n - 1$. So the sum of the degree of all nodes together in both graph is equal to $n(n-1)$

(c) Note that $n(n-1) = 2S$ and S is even. So $4 \mid (n)(n-1)$. Also note that between n and $n-1$ one is odd and one is even. So either $4 \mid n$ or $4 \mid n-1$.

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